Differentiable transmon design with a tensor-native finite-element electromagnetics solver: gradient-based optimization of charging energy, cross-validated against Palace

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Abstract

The SingleTransmon example of DeviceLayout.jl publishes the qubit and resonator frequencies of an initial and an optimized transmon design, computed with Palace at DeviceLayout's own second-order discretization. We cross-check those numbers with two independently implemented finite-element solvers, GEODE-FEM and Palace. Palace at DeviceLayout's discretization reproduces both published pairs to within 0.7%, and on shared meshes the two solvers agree on the qubit mode to 10^{-4}%. Under refinement both converge to qubit frequencies 7.3% and 8.7% above the published 4.14 and 3.005 GHz, and to resonator frequencies 2.7% above, so the published values are limited by discretization. A static route agrees: a certified two-sided bracket on the converged island capacitance with the sapphire tensor, $[79.56, 79.76]$ fF, read through a two-node circuit of junction and resonator, puts the initial qubit at 4.45 GHz and caps it at 4.477 GHz. We then repeat DeviceLayout's two-parameter optimization with Newton steps whose Jacobian comes from GEODE-FEM's shape derivative of the eigenfrequencies. At DeviceLayout's discretization it meets the published stopping criterion in 3 Palace runs, where the derivative-free optimizer used 9. Aimed at converged frequencies, it finds that the 3.0/4.0 GHz design needs a capacitor about 19% longer than the published one (16–38% within the extrapolation band) and a resonator 2.4% longer. Converged frequencies are extrapolations with stated bands, not bounds; the capacitance brackets are certified for the meshed polygonal geometry.

1 Introduction

Superconducting-qubit layouts are designed by iterating field solves: a designer parameterizes a geometry, simulates it, compares the extracted frequencies or Hamiltonian parameters with targets, and adjusts. The state-of-practice automated workflow, QDesignOptimizer (Eriksson et al., 2025), makes the loop explicit and steers it with analytic models because its field solver returns observables but no derivatives of them. Open tools now cover the whole loop. DeviceLayout.jl (AWS Center for Quantum Computing, 2025a) builds the layout and its mesh, and Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based solver (Anderson et al., 2021; Andrej et al., 2024), computes the eigenmodes. DeviceLayout's SingleTransmon example (AWS Center for Quantum Computing, 2025b; Peairs and Carson, 2025) runs the loop end to end. It publishes an initial transmon-plus-resonator design at 4.14 and 5.591 GHz and a design optimized to targets of 3.0 and 4.0 GHz, at 3.005 and 3.982 GHz, found with a derivative-free optimizer in nine Palace runs.

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Those frequencies come from one discretization, and the example itself warns that they are sensitive to it. This paper measures how sensitive. Our main result is that the published frequencies are discretization-limited: two independently implemented solvers converge to qubit frequencies 7–9% above them, a certified bound on the island capacitance gives the same shift by a static route, and a gradient-assisted re-optimization at converged resolution finds that the optimized design needs a capacitor about 19% longer.

The second solver is GEODE-FEM ("geode-fem"), a Rust finite-element code on the Burn tensor framework (Tracel AI, 2024). It solves the same $H(\mathrm{curl})$ eigenproblem as Palace with an independent implementation, and it computes design derivatives: shape and material sensitivities of its electrostatic and driven operators, and the Hellmann–Feynman shape derivative of an eigenfrequency. On the committed DeviceLayout mesh, given as the same input file to both codes, geode-fem reproduces Palace's six eigenmodes above 4.5 GHz to $\leq 0.033\%$ and the qubit mode to 5×10^{-5} % (Section 4). This parity is *a credential, not the contribution* — matching Palace is the price of entry — and it is, incidentally, the first independent third-party validation of the Palace/DeviceLayout.jl transmon stack, which has no publication of its own.

Our contributions are:

- **A converged cross-check of a published design example.** Both published geometries are meshed by DeviceLayout itself and solved by both codes on a resolution ladder with first- and second-order elements. Palace at DeviceLayout's discretization reproduces the published pairs to under 0.7%; the converged estimates lie 7.3% and 8.7% above the published qubit frequencies and 2.7% above the resonators, outside every extrapolation band (Section 5).

- **Certified capacitance brackets with the sapphire tensor.** A conforming upper bound and an equilibrated-flux lower bound, a classical pairing applied here to an island capacitance with a floating feedline, bracket the converged island capacitance of the unmodified device in $[79.56, 79.76]$ fF. Read through the junction and resonator circuit they give the initial qubit at 4.45 GHz, inside the eigenmode band, and a cap of 4.477 GHz. The certified difference between the sapphire tensor and its scalar average is 2.5–3.0% (Section 6).

- **A gradient-assisted re-optimization.** Newton steps in DeviceLayout's own variables, with a Jacobian from geode-fem's eigenfrequency shape derivative and a transmission-line model, meet DeviceLayout's criterion in 3 Palace runs against the published 9, and at converged resolution move the optimized capacitor from 1217.678 to about 1446 μm (Section 7).

- **A validated sensitivity layer.** Material and geometry derivatives of the electrostatic and driven $H(\mathrm{curl})$ operators, each checked against finite differences of the full pipeline with mutation tests (Section 8).

- **The correctness credential.** Same-input-mesh agreement with Palace on six eigenmodes and the qubit mode, with the 17.49 GHz mode both codes share identified as an artifact of the distributed port model, doubling as the first independent validation of the Palace/DeviceLayout.jl stack.

Section 2 places the work. Section 3 describes the solvers, the device and the models; Section 4 the same-mesh agreement; Sections 5–7 the cross-check, the certified capacitance and the re-optimization; Section 8 the sensitivity layer; and Section 10 what is and is not established. Appendix A reports an electrostatic design loop with a certified acceptance test, and Appendix B the performance context.

2 Related Work

Design sensitivities in finite-element electromagnetics. Adjoint and direct sensitivity analysis of frequency-domain electromagnetic solvers is an established field. Nikolova et al. (2004) review adjoint-variable methods for high-frequency structures across FEM, MoM, TLM and FDTD. On tetrahedral vector elements, Webb and coworkers derived design sensitivities with high-order vector elements and of the frequency response, and used them to optimize microwave devices (Nair and Webb, 2003; Webb, 2001, 2002), and continuum shape sensitivity for high-frequency devices has been formulated as well (Choi et al., 2011). Shape adjoints for Maxwell cavity eigenvalues were developed for accelerator cavities (Akçelik et al., 2005, 2008), and a recent study optimizes Maxwell eigenvalues by shape variations with hand-derived adjoint gradients (Herter et al., 2026). Commercial solvers expose derivatives: HFSS computes analytic S-parameter derivatives with respect to geometric and material variables by reusing the forward LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL offers forward and adjoint sensitivities and electromagnetic shape optimization (COMSOL AB, 2023; Jensen, 2022). Automated shape differentiation for general FEM exists in UFL/Firedrake (Ham et al., 2019) and NGSolve (Gangl et al., 2021). Palace exposes no derivatives. For capacitance, adjoint sensitivities are standard in interconnect extraction (El-Moselhy et al., 2008) and FEM (Ren et al., 2012), and the result that the shape derivative of a stored-energy capacitance reduces to its explicit term is the classical self-adjoint case (Park, 2019). What geode-fem adds is an open implementation in which these derivatives come from the same element kernels as the solve, applied here to a qubit design loop; the methods are not new.

Superconducting-qubit design optimization. QDesignOptimizer (Eriksson et al., 2025) wraps non-differentiable simulation with physics-guided updates. The SQcircuit line (Rajabzadeh et al., 2023, 2024) and SuperGrad (Wang et al., 2025) differentiate at the circuit-Hamiltonian level, so the map from geometry to circuit parameters still runs through field simulation. SQuADDS (Shanto et al., 2024) replaces simulation by lookup in a validated design database, and Eun et al. (2023) shape-optimize transmon pads with FEM and a derivative-free global optimizer. Qiskit Metal's quasi-lumped methods (Minev et al., 2021b) and energy-participation and black-box quantization (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) supply the parameter extraction. To our knowledge, within qubit EM design no prior work drives a layout with gradients from the field solve.

Certified two-sided capacitance bounds. Bracketing a capacitance between complementary energy principles is classical, from the analytic variational bounds of Polya and Szegö (1951) to finite-element dual formulations (Hammond and Penman, 1976; Penman and Fraser, 1983), and in three dimensions Specogna (2011) gives complementary formulations that bound capacitance from both sides. Our lower bound uses the hypercircle idea of Prager and Synge (1947) with Raviart–Thomas fluxes (Raviart and Thomas, 1977) equilibrated patch by patch (Braess and Schöberl, 2007; Braess et al., 2009; Ern and Vohralík, 2015), the route of guaranteed goal-oriented bounds such as Parés et al. (2021). The method is not new. What we add is its application to an island capacitance with a floating conductor on a production qubit layout, used as a design-acceptance test; we found no certified capacitance bound in the qubit-design literature.

Cross-checks of Palace. SQDMetal (Sommers et al., 2025) cross-checks a Palace workflow against COMSOL and Ansys with mesh-convergence studies, and Ye et al. (2025) compare a Palace workflow with cryogenic measurement at the $\sim$0.3% level. Both use models built independently for each tool. Our comparison instead holds the input mesh and the junction model fixed and changes the solver implementation, and it adds a converged reference for a published design. In

V&V terms (Oberkampf and Roy, 2010; Roache, 1997) it is code verification and code-to-code comparison, not validation against measurement. Code-to-code benchmarking on precisely specified problems is a long tradition in computational electromagnetics (Hoffmann et al., 2009). Palace and DeviceLayout.jl have no methods publication (AWS Center for Quantum Computing, 2025a; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson, 2025).

Differentiable FEM on general tensor stacks. Differentiable electromagnetics is most developed in photonics, on structured grids and integral methods (Hammond et al., 2022; Hughes et al., 2019; Kim et al., 2024; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022). Differentiable FEM on autodiff tensor stacks exists outside EM (Liang et al., 2023; Xue et al., 2023) and for general Galerkin assembly (Chi and Wen, 2026; Wen et al., 2026). FEM kernel generation is a long tradition (Alnæs et al., 2014; Homolya et al., 2018; Kirby and Logg, 2006; Rathgeber et al., 2017), and its GPU form is libCEED (Brown et al., 2021), on which Palace runs. geode-fem phrases assembly as batched tensor algebra on a general-purpose ML stack (Chen et al., 2018; Ragan-Kelley et al., 2013) so that element-level derivatives are available by construction.

3 Solvers, Device and Models

3.1 geode-fem

geode-fem assembles first-order Nédélec (Nédélec, 1980) and scalar P1/P2 element operators as batched $[n_{\text{elem}}, k, k]$ tensor operations on Burn, whose cubecl compiler targets CPU, CUDA and WebGPU from one source. Its eigensolver is shift-invert Lanczos (Lehoucq et al., 1998) on a real symmetric pencil, factorized with faer (Kazdadi, 2026); large electrostatic solves use conjugate gradients with an algebraic-multigrid preconditioner in native f64. Autodiff reaches everything up to the linear solve. The sparse factorization is outside the tape, so a discrete-adjoint layer supplies the missing derivative with one transpose solve on the same factorization (Section 8). The eigenfrequency derivative used in Section 7 needs no adjoint solve, because the pencil is symmetric: for a simple eigenpair (λ, x) with $x^{\mathsf{T}} M x = 1$,

$$\frac{d\lambda}{d\theta} \;=\; x^{\mathsf{T}}\Big(\frac{\partial K}{\partial \theta} - \lambda\,\frac{\partial M}{\partial \theta}\Big)x, \tag{1}$$

the Hellmann–Feynman (Nelson) formula (Nelson, 1976). geode-fem evaluates the element derivatives $\partial K/\partial\theta$ and $\partial M/\partial\theta$ by forward-mode dual numbers through the same tensor-permittivity element kernels it assembles, along a node velocity $\partial X/\partial\theta$.

3.2 Device, mesh and materials

The device is DeviceLayout.jl v1.15.0's SingleTransmon example (AWS Center for Quantum Computing, 2025a): a transmon with a meander readout resonator and a feedline on sapphire, in a vacuum box (Figure 1). Its two design parameters are the transmon's `capacitor_length` c and the resonator's `total_length` L; the published initial design has $c = 620\,\mu$m and $L = 5000\,\mu$m. The sapphire permittivity is $\mathrm{diag}(9.3, 9.3, 11.5)$ with the optic axis normal to the chip. The configuration rotates the material axes about the chip normal by about 36.87°; for a uniaxial tensor whose optic axis is that normal the rotation is the identity, so the lab tensor is diagonal. Where a scalar model is used it is the trace average $\bar{\varepsilon} = 10.03$.

Two mesh families appear. The committed fixture is DeviceLayout's mesh of the initial design (22,684 nodes, 133,314 tetrahedra), pinned by SHA-256; Section 4 and the capacitance brackets of

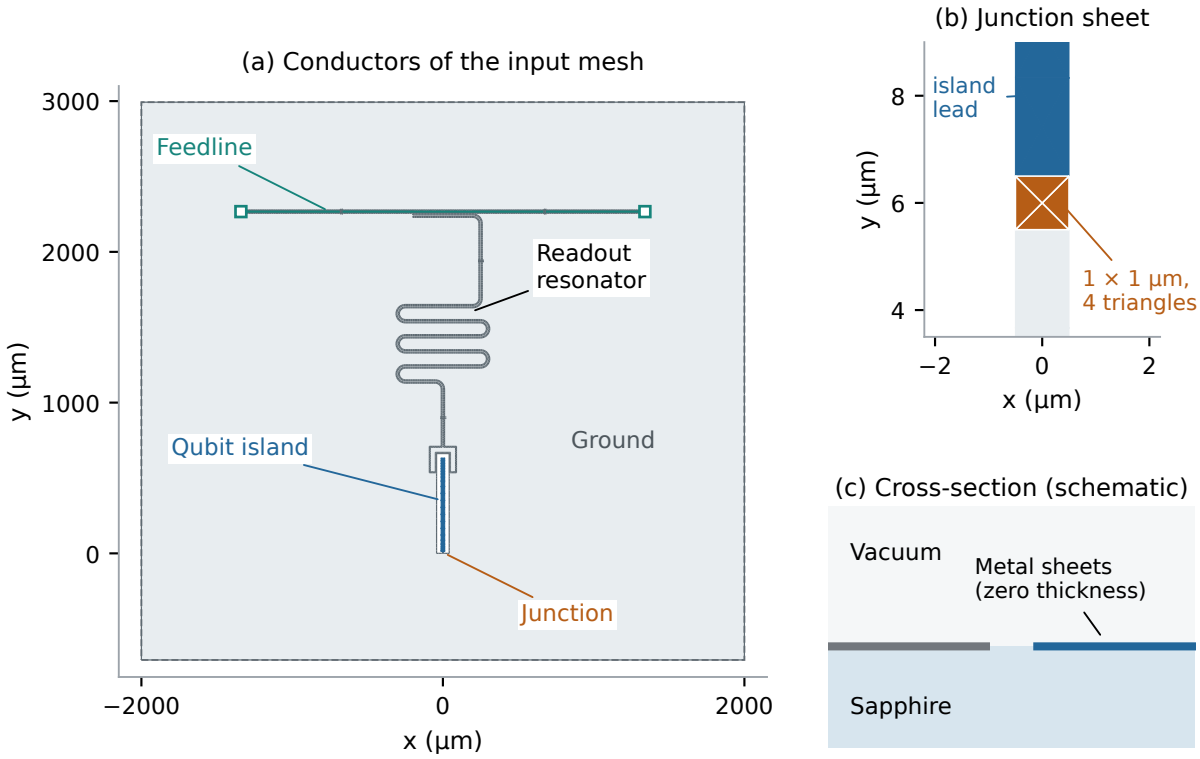


Figure 1: Benchmark geometry, drawn from the committed SHA-256-pinned MSH 4.1 input mesh of the DeviceLayout.jl v1.15.0 `SingleTransmon` device. (a) Top view of the three node-disjoint metal conductors of Section 6: the feedline (teal), the qubit island (blue), and the ground plane together with the meander readout resonator (gray), whose etched gaps are drawn as dark outlines. Squares mark the two readout ports; the dashed line is the exterior-boundary outline. (b) The $1 \times 1\,\mu$m junction `lumped_element` sheet with its four-triangle triangulation. (c) Material cross-section, schematic and not to scale; the `substrate` and `vacuum` volume groups fill the enclosing box. Both solvers read this file (22,684 nodes; 133,314 tetrahedra); Palace then refines it internally around the metal sheets (Section 4).

Section 6 use it and its refinements, which leave its polygonal surfaces unchanged. For the cross-check and the re-optimization, DeviceLayout itself meshes each geometry with the two parameters exposed, at first order for the ladders and at its own second order for the runs that reproduce its discretization. On the initial geometry the two families give the same Palace qubit at h (3.4527 against 3.4520 GHz). PEC is applied on the metal sheets and the box; the readout ports are left open in every eigenproblem except the DeviceLayout-configuration runs of Section 5.

3.3 Junction and circuit models

The Josephson junction is a linear lumped element, $L_J = 14.860\,\text{nH}$ in parallel with $C_J = 5.5\,\text{fF}$, on the four-triangle sheet Γ of Figure 1b, exactly as in DeviceLayout's Palace configuration. With port length ℓ and width w the uniform lumped-port construction adds

$$K_{\text{port}} = \frac{\ell}{w\,L_J}\,S_\Gamma \qquad \text{and} \qquad M_{\text{port}} = \frac{C_J\,\ell}{w}\,S_\Gamma \tag{2}$$

Table 1: Eigenmode agreement from the identical input mesh (22,684 nodes, 133,314 tetrahedra), first order in both solvers, junction as the lumped $L_J \parallel C_J$ shunt. geode-fem solves 156,863 Nédélec unknowns (133,108 interior); Palace refines the mesh internally around the metal sheets and solves 183,891. geode-fem frequencies are the committed three decimals (six for the qubit), and Δ is the committed relative difference. These are same-mesh values, not converged frequencies. Source: `benchmarks/transmon_eigen/results.toml`.

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
qubit	transmon qubit	3.452751	3.452749306805	0.00005
resonator	readout resonator	5.153	5.151335830348	0.032
mode 2	box/cavity	15.465	15.46052107794	0.029
port sheet	port-sheet artifact	17.490	17.49010903536	0.001
mode 4	box/cavity	18.693	18.69165792915	0.007
mode 5	box/cavity	20.703	20.69755679425	0.026
mode 6	box/cavity	26.088	26.08089940472	0.027

to the pencil $(K + K_{\text{port}})x = \omega^2(M + M_{\text{port}})x$, where S_Γ is the isotropic surface matrix on Γ. Palace's `LumpedPort` adds the same terms with the same scalings, so the default models of the two codes coincide. The isotropic sheet also stores energy in a zero-voltage field pattern confined to the patch, which produces a mode at 17.49 GHz in both codes (Section 4). geode-fem's rank-1 port-voltage shunt replaces S_Γ by $b\,b^{\mathsf{T}}$, with $b_i = (1/w)\int_\Gamma N_i \cdot \hat{e}\, dS$ the port-voltage functional. It removes that mode and keeps the others to four decimals on the fixture, and it is the junction model of every geode-fem eigensolve in Sections 5 and 7.

The electrostatic model splits the metal into three conductors: ground with the resonator, the floating feedline, and the island. Its island capacitance with the feedline floating and neutral, $C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$, excludes C_J. A two-node circuit connects it to the eigenproblem: a bare qubit $1/(2\pi\sqrt{(L_J + L_s)(C_\Sigma + C_J)})$, with L_s the series inductance of the junction's current path, coupled with coefficient k to the resonator. On the fixture L_s and k were measured on three and two discretizations; extrapolated, they give bands of 0.113–0.125 nH and 0.0227–0.0310, which are estimates, not bounds.

4 Same-Input-Mesh Agreement With Palace

Both codes read the committed fixture with the same materials and junction model at first order. Table 1 gives the result. Palace returns the qubit mode only when its search target lies below it, because its eigensolver searches above the target (read from Palace's source and configuration schema); the six modes above 4.5 GHz come from a run with `Target` = 4.5 GHz and the qubit from one with `Target` = 2 GHz. All six agree to $\leq 0.033\%$ (worst case 0.032%), and the qubit mode to 5×10^{-5} %, with port participation 0.9942 in geode-fem and port EPR 0.99424 in Palace. Doubling L_J moves the qubit by a ratio of 0.70810 in Palace and 0.7081 in geode-fem, close to $1/\sqrt{2}$, and a Palace rerun from the committed configuration reproduces its `eig.csv` bit for bit.

The 17.49 GHz mode needs care, and its explanation below is an inference that a synthetic test supports. In geode-fem all its stiffness energy is in the port term ($p = 1.000$), while Palace, which builds its port EPR from the port voltage, gives it 2.49×10^{-8}; its energy sits in a surface pattern with zero net voltage. The junction patch has one free node, whose hat-function gradient carries no port voltage but still stores energy in the isotropic sheet term, with Rayleigh quotient $1/(L_J C_J)$, or $f_{LC} = 17.60$ GHz, lowered to 17.49 GHz by the adjacent volume mass. A two-terminal junction cannot carry that pattern. Its 0.001% agreement therefore tests that the two codes assemble the same port operator and nothing more, and the rank-1 shunt of Section 3.3 removes the mode.

Table 2: Published frequencies, Palace at DeviceLayout's discretization, and the converged estimates with their extrapolation bands, in GHz. Percentages are relative to the published value. The estimates and bands are extrapolations, not bounds. Source: `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

	Published	At DeviceLayout's discretization	Converged estimate	Band
Initial qubit	4.14	4.1683 (+0.68%)	4.444 (+7.3%)	$[4.415, 4.677]^*$
Initial resonator	5.591	5.6132 (+0.40%)	5.740 (+2.7%)	$[5.723, 5.817]$
Final qubit	3.005	3.0164 (+0.38%)	3.265 (+8.7%)	$[3.236, 3.471]$
Final resonator	3.982	3.9956 (+0.34%)	4.088 (+2.7%)	$[4.077, 4.139]$

*Capped at 4.477 GHz by the static route of Section 6: $[4.415, 4.477]$.

The fixture is coarse. Palace's own refinements of it move the qubit from 3.4527 GHz to 3.9528 GHz at $h/2$ (+14.5%) and to 4.1721 GHz at second order on h (+20.8%), and the resonator by +6.1% at $h/2$. The agreement above is code verification on one discretization; Section 5 supplies the converged numbers.

5 The DeviceLayout Cross-Check

The published designs. The DeviceLayout.jl blog (Peairs and Carson, 2025) reports the initial design ($c = 620\,\mu$m, $L = 5000\,\mu$m) at $[4.14, 5.591]$ GHz and the final design ($c = 1217.678\,\mu$m, $L = 7243.955\,\mu$m) at $[3.005, 3.982]$ GHz, reached by the derivative-free PRIMA optimizer in nine Palace runs "within 1% of our target frequencies". The example source (AWS Center for Quantum Computing, 2025b) sets the targets to $[3.0, 4.0]$ GHz, works in the variables $x_1 = \sqrt{620\,\mu\text{m}/c}$ and $x_2 = 5000\,\mu\text{m}/L$, and stops when $\text{err} = \frac{1}{2}\sum_m (f_m - t_m)^2/t_m^2 < 10^{-4}$. Its frequencies are computed with Palace on DeviceLayout's mesh at mesh order 2 and solver order 2, with a first-order absorbing far field and two 50 Ω readout ports.

Reproduction at DeviceLayout's discretization. Palace, run on DeviceLayout's second-order mesh with the example's configuration file, gives $[4.1683, 5.6132]$ GHz for the initial design and $[3.0164, 3.9956]$ GHz for the final one, 0.68% and 0.40% above the published initial pair and 0.38% and 0.34% above the final pair. The same cells with the PEC-box model used everywhere else differ from the configuration-file runs by at most 0.003%, so the boundary model does not matter here.

The ladder. For each geometry, Palace runs at order 1 on DeviceLayout's first-order mesh h and its uniform refinements $h/2$ and $h/4$ (up to 12.6 million unknowns), and at order 2 on h and $h/2$; geode-fem runs at order 1 on h and $h/2$, on the same meshes. Every series rises under refinement (Figure 2). The sheet-edge singularity of the field, $E \sim r^{-1/2}$, limits the asymptotic eigenvalue rate on uniform meshes to $O(h)$ at any element order; the order-1 ladders show pre-asymptotic orders of 0.61 (qubit) and 0.70–0.73 (resonator).

Extrapolation rule. The *estimate* is the mean of two extrapolations at the asymptotic order 1 in the frequency, $2f_2 - f_1$: one from the order-1 pair ($h/2$, $h/4$) and one from the order-2 pair (h, $h/2$). The *band* spans six values: those two; the same pairs extrapolated at order 1 in the eigenvalue variable $\lambda \propto f^2$, $\sqrt{2f_2^2 - f_1^2}$, which set the low side; and the same pairs at the observed order, which set the high side. The band is an extrapolation range, not a bound, and its high side is kept wide on purpose. For the initial qubit, the static route of Section 6 caps the band at 4.477 GHz.

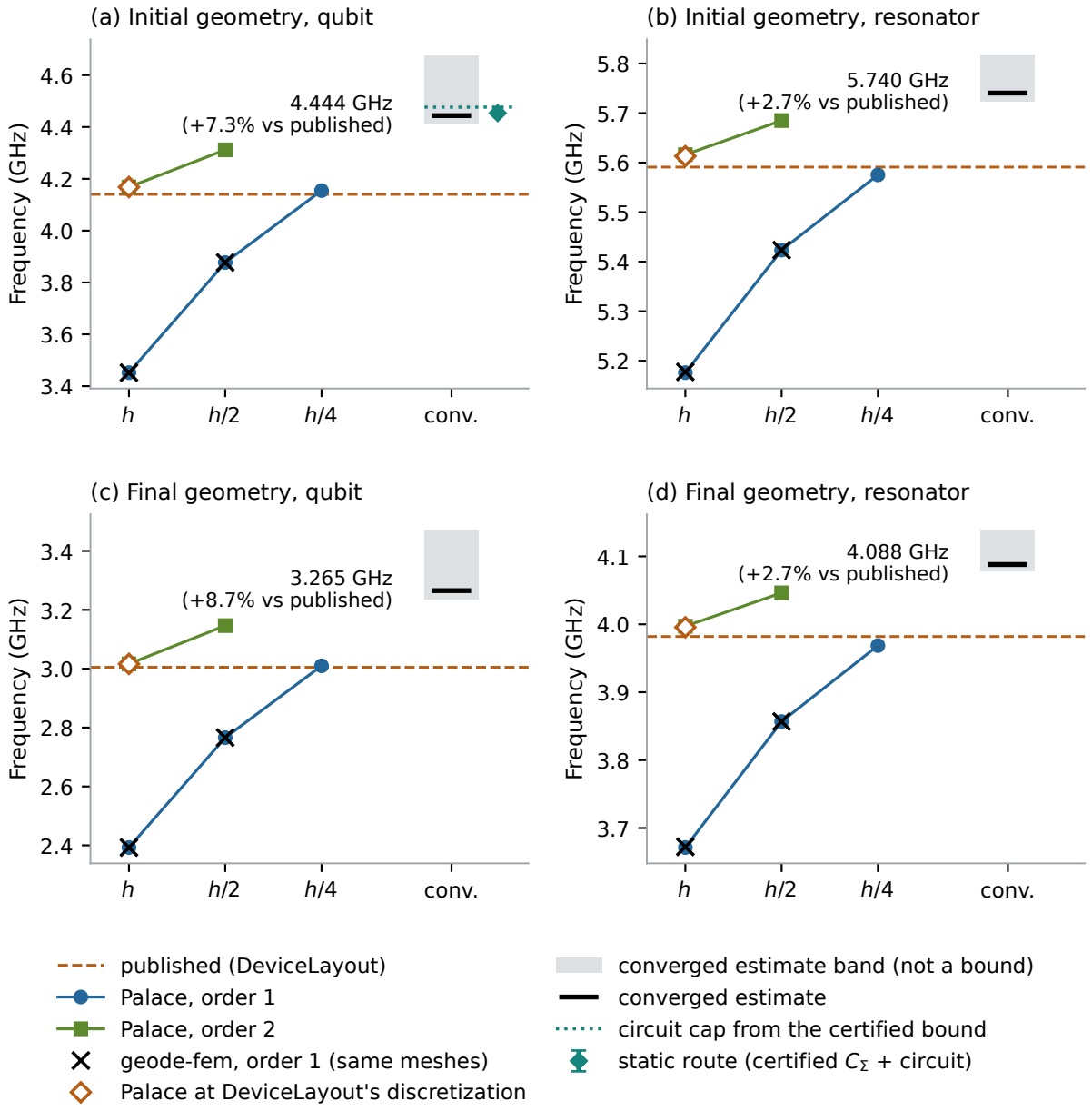


Figure 2: Resolution ladders of the two published DeviceLayout.jl SingleTransmon designs. Palace at order 1 on DeviceLayout's mesh h and its uniform refinements, Palace at order 2, and geode-fem at order 1 on the same meshes (crosses, on top of Palace's order-1 points); the open diamond is Palace on DeviceLayout's own second-order mesh with its configuration file, the discretization of the published values (dashed). At the right, the converged estimate with its extrapolation band, which is not a bound. In (a), the static route of Section 6 (certified capacitance bracket read through the junction–resonator circuit) and the circuit cap from the certified lower bound. Source: `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

Result. Table 2 gives the converged estimates. The initial qubit converges to 4.444 GHz and the final qubit to 3.265 GHz, 7.3% and 8.7% above the published values; both resonators converge 2.7% above theirs. Every published value lies below its band. Measured against the estimates, DeviceLayout's discretization sits 6.2% and 7.6% low on the two qubits and 2.2% and 2.3% low on the resonators, and the first-order mesh h alone sits 22.3% and 26.7% low on the qubits.

Same-mesh agreement on the ladder. geode-fem and Palace agree on the qubit to 10^{-4} % at every shared level of both geometries. On the resonator they differ by 0.03% at h, where Palace adds 4,946 (initial) and 5,944 (final) elements by local bisection of interior boundary tetrahedra, and by 0.0000% at $h/2$, where no bisection occurs and both codes solve the same discrete problem. Rerunning one cell with AMS in place of SuperLU, or feeding geode-fem a format round-trip of the mesh, reproduces the same values.

Scope. The converged values are estimates for DeviceLayout's geometry under the PEC-box, linear-junction model, from one mesh family per geometry refined uniformly; the band does not include mesh-family spread. They are not a statement about a fabricated device.

6 A Static Route: Certified Island Capacitance

The certificate. For the reduced electrostatic problem (island at potential 1, ground and resonator at 0, feedline floating with zero net charge), the potential ϕ satisfies $0 \le \phi \le 1$ by the maximum principle, and $C_\Sigma = \int \varepsilon \nabla\phi \cdot \nabla\phi$ with $\varepsilon = \varepsilon_0\varepsilon_r$. Every conforming finite-element energy is an upper bound on C_Σ, because C_Σ is the minimum of the energy over admissible potentials. For the lower bound, take any piecewise polynomial field D on the mesh cut along the zero-thickness sheets. Integrating by parts and using $0 \le \phi \le 1$ on every term gives

$$\int D \cdot \nabla\phi \;\ge\; N(D) \;=\; Q_{\text{island}}(D) - |Q_{\text{feed}}(D)| - \|\operatorname{div} D\|_{L^1} - \|[D \cdot n]\|_{L^1(\mathcal{F})}, \tag{3}$$

where Q are the fluxes of D into the two conductors and $[D \cdot n]$ is the normal jump on the interior faces $\mathcal{F}$ off the sheets. Cauchy–Schwarz, $\left(\int D \cdot \nabla\phi\right)^2 \le \int D \cdot \varepsilon^{-1} D \int \varepsilon \nabla\phi \cdot \nabla\phi$, then gives, whenever $N(D) > 0$,

$$C_\Sigma \;\ge\; \frac{N(D)^2}{\int D \cdot \varepsilon^{-1} D}. \tag{4}$$

The bound holds for any D, so solver tolerance can only weaken it. We take D from the P2 potential, equilibrated patch by patch into Raviart–Thomas fluxes of order one to three, with an independent Crouzeix–Raviart flux as a check. The construction is validated against an exact parallel plate, an analytic coaxial bracket and closed-form checks of each term, with mutation tests that reject a grounded-feedline or jump-free certificate. *Certified* in this paper means rigorous for the discrete polygonal geometry of the mesh, with lower bounds floored and upper bounds ceiled. The certificate is evaluated in plain f64, not interval arithmetic; the worst-case summation error is about 10^{-10} relative, far below the rounding. Refinement does not move the polygonal surfaces, so all meshes of one geometry bound the same converged value, and a family bracket is the largest lower bound and the smallest upper bound.

The unmodified device. Uniform refinement of the fixture with P1 and P2 elements gives the upper bounds of Table 3. On adaptively refined meshes the certificate closes the gap: the converged island capacitance of the unmodified initial design X^0 lies in $[77.40, 77.61]$ fF with the scalar $\bar{\varepsilon}$ and

Table 3: The island capacitance C_Σ (fF) of the unmodified device X^0 on uniform refinements of the committed mesh, and the certified brackets of its converged value from adapted meshes. Each uniform entry is a conforming energy and therefore an upper bound. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml` in `benchmarks/transmon_diffopt/`.

Level	Tetrahedra	P1, scalar	P2, scalar	P1, tensor	P2, tensor
h	133,314	137.706752	90.044141	136.684731	91.717846
$h/2$	1,066,512	101.359397	83.015285	102.375301	85.008425
$h/4$	8,532,096	87.753422	n/a	89.493342	n/a
Certified converged value		$[77.40, 77.61]$		$[79.56, 79.76]$	

in $[79.56, 79.76]$ fF with the sapphire tensor (Figure 3). The certified tensor-minus-scalar difference is 1.95–2.35 fF, or 2.5–3.0% of the scalar value, and the same relative range holds for the two designs of Appendix A. The committed mesh overstates the tensor C_Σ by at least 56.92 fF, which is at least 71.3% of the converged value, and the scalar one by at least 77.4%.

From capacitance to the qubit frequency. Reading the certified tensor bracket through the two-node circuit of Section 3.3, over every corner of the C_Σ, L_s and k bands, with the resonator at its converged estimate, gives the initial dressed qubit in $[4.449, 4.458]$ GHz. That range lies inside the eigenmode band of Section 5, and the eigenmode estimate is 0.22% below its midpoint. The range is an estimate, because L_s and k are. One statement is sharper. A series inductance and a coupling to a higher mode can only lower the qubit, so under the circuit model the certified lower bound $C_\Sigma \geq 79.56$ fF caps it at $1/(2\pi\sqrt{L_J(C_\Sigma + C_J)}) = 4.477$ GHz. The cap trims the high side of the eigenmode band, which comes from extrapolation at the observed pre-asymptotic order (4.677 GHz). Two caveats apply: the bracket is certified on the fixture, which an earlier generator produced from the same DeviceLayout geometry, not on the meshes of the ladder; and the circuit is a model of the full-wave problem.

7 Gradient-Assisted Re-Optimization

The cross-check says that the published final design does not meet its targets at converged resolution. We repeat DeviceLayout's optimization twice: once against its own discretization, to compare cost with the published run, and once against converged frequencies.

The capacitor_length gradient. DeviceLayout keeps the junction fixed and builds the island upward from it, so a change of c translates every node above the island rigidly, stretches the island band and re-centres the box. geode-fem turns this motion into a node velocity $\partial X/\partial c$, checks on the mesh that every metal outline node in a stretched or blended band lies on an edge the motion maps onto itself (3,146 nodes on the initial mesh), and evaluates Eq. (1) with the sapphire tensor for both modes. The contraction takes 0.44 s. On the initial mesh it matches a central difference of two re-solves of the moved mesh ($\pm 0.5\,\mu$m) to every printed digit: -2.443227×10^{-3} GHz/μm for the qubit and -3.964×10^{-6} GHz/μm for the resonator.

The total_length column. DeviceLayout re-lays the meander when L changes, so no fixed-topology node motion represents it. A transmission-line model holds $f_r(L + \delta)$ constant, with $\delta = 216\,\mu$m from one finite difference of two DeviceLayout meshes ($L = 5000$ and $5250\,\mu$m), and takes $\partial f_q/\partial L = 0$. δ is weakly determined: remesh noise of about 0.15% in f_r is roughly $\pm 150\,\mu$m

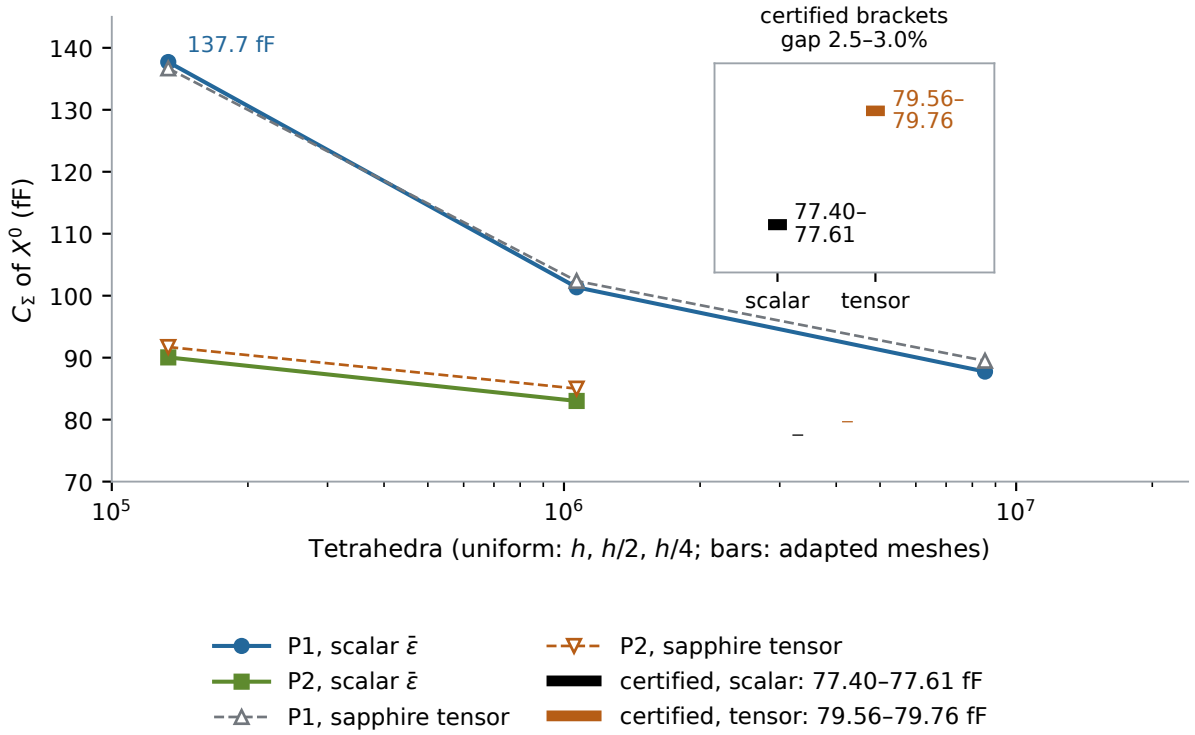


Figure 3: Mesh convergence of the island capacitance C_Σ of the unmodified device under uniform refinement of the committed mesh (Table 3), with the scalar trace-averaged permittivity and with the sapphire tensor. Every marker is a conforming upper bound on the converged value of the committed polygonal geometry. The bars are the certified two-sided brackets from adapted meshes, drawn at the size of the mesh that sets them and magnified in the inset; the inset title gives the certified tensor-minus-scalar gap. Sources: `c_sigma_convergence.toml`, `c_sigma_bracket.toml` and `c_sigma_bracket_tensor.toml`.

Table 4: Newton iterates against DeviceLayout's discretization (Palace on its second-order mesh with its configuration file). Step 0 is the initial design. err is DeviceLayout's measure against $[3.0, 4.0]$ GHz; its criterion is 10^{-4}. Source: `benchmarks/transmon_devicelayout_optimize/results.toml`.

Step	c (μm)	L (μm)	f_q (GHz)	f_r (GHz)	err
0	620.000	5000.000	4.1683	5.6132	1.6×10^{-1}
1	1338.338	7137.638	2.8859	4.0465	7.9×10^{-4}
2	1225.346	7223.250	3.0120	4.0009	8.0×10^{-6}
3	1236.360	7224.900	3.0013	4.0011	1.4×10^{-7}

in δ, and a second estimate gives about 363 μm. The model enters only the Jacobian, and every iterate is evaluated on the true objective, so a poorer Jacobian costs steps, not accuracy.

Newton on the high-fidelity objective. DeviceLayout re-meshes every design, so the first-order objective is not smooth in the parameters: from $c = 620$ to 650 μm the remeshed finite difference of the qubit frequency is -5.23×10^{-3} GHz/μm, against a fixed-topology derivative of -2.28×10^{-3}. The iteration therefore runs Newton directly on the high-fidelity objective F, in DeviceLayout's variables (x_1, x_2), in which the frequencies are close to linear. The c column of its Jacobian is geode-fem's first-order derivative scaled by the measured fidelity ratio $S = F/f_h$ at the same design. Each step costs one geode-fem first-order solve with its gradient (19–25 s) and one high-fidelity evaluation; the one-time δ calibration is setup and is not counted.

Against DeviceLayout's discretization. Table 4 lists the iterates. The run meets DeviceLayout's criterion at step 2, after 3 Palace runs counting the initial one as DeviceLayout does, against PRIMA's 9; the three geode-fem solves add about 70 s, against about 4 min per Palace run on 32 ranks (Figure 4b). Step 3 reaches err $= 1.4 \times 10^{-7}$ at $(1236.360, 7224.900)$ μm, +18.7 and −19.1 μm from the published $(1217.678, 7243.955)$ μm. About +10 μm of the c offset is the Palace build: ours gives $[3.0164, 3.9956]$ GHz at the published final design, against the published $[3.005, 3.982]$ GHz. The rest is where PRIMA stopped; its published pair sits 0.17% and 0.45% from the targets, inside its 1% rule. The comparison is with PRIMA's published run count, not with its trajectory, which is not published.

Against the converged frequencies. Starting from the published final design, whose converged estimate is 3.265 and 4.088 GHz, each Newton step uses the converged estimate and the geode-fem gradient at the current design, and each new design is evaluated by a full ladder (Palace at order 1 on h, $h/2$, $h/4$ and order 2 on h, $h/2$, geode-fem at order 1 on h, $h/2$) with the rule of Section 5. Table 5 gives the two steps. The first overshoots the qubit by −1.09%; the second lands at +0.16% and −0.04% of the targets, with both targets inside the bands. On both ladders geode-fem and Palace agree on the qubit to 2×10^{-4} %.

The converged design. At converged resolution the 3.0/4.0 GHz design needs a substantially longer capacitor than the published one: the point estimate is $c \approx 1446$ μm, +19% (+228 μm), with a resonator 2.4% longer (+172 μm). Mapped linearly through $dF/dc \approx S\,\partial f_h/\partial c = -9.05 \times 10^{-4}$ GHz/μm, the step-2 qubit band corresponds to c between about 1418 and 1685 μm, +16% to +38%. The point estimate probably leans low: the order-1 and order-2 series share a limit at order 0.87 (qubit) and 0.84 (resonator), which puts step 2 at 3.039 and 4.019 GHz, +1.3% and +0.5% over the targets, so the converged design is probably at or above the point-estimate c. Step 2 therefore

Table 5: Newton steps against the converged frequencies. Each design is evaluated by a full resolution ladder; the converged values are estimates with extrapolation bands, not bounds. Source: `benchmarks/transmon_devicelayout_optimize/results.toml` and, for the published final design, `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

Design	c (μm)	L (μm)	f_q [band] (GHz)	f_r [band] (GHz)
Published final	1217.678	7243.955	3.265 [3.236, 3.471]	4.088 [4.077, 4.139]
Step 1	1481.657	7407.945	2.967 [2.938, 3.188]	4.004 [3.993, 4.055]
Step 2	1446.078	7416.352	3.005 [2.979, 3.221]	3.998 [3.987, 4.067]

Table 6: The discrete-adjoint sensitivity layer. Each cell: one forward and one adjoint solve on one factorization, exact element Jacobians, and the observed agreement with a central finite difference of the full pipeline. Modules in `crates/geode-core/src/`.

Operator	Parameter	Module	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\varepsilon$	`adjoint`	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	`shape`	$\sim 1 \times 10^{-9}$
driven H(curl)	material $\partial/\partial\varepsilon$	`driven::adjoint`	2.3×10^{-5} (worst region)
driven H(curl)	geometry $\partial/\partial X$	`driven::shape`	2.2×10^{-9}, 1.2×10^{-8} (two maps)

lands on the targets within the band; it is not a verified hit of 3.0/4.0 GHz at convergence. Each ladder took 1.4 and 1.6 h of Palace wall time, dominated by the order-2 $h/2$ cell.

8 The Sensitivity Layer

For a solve $A(\varepsilon, X)\, x = b(X)$ and an observable $g(x)$, the discrete adjoint $A^{\mathsf{T}}\lambda = \partial g/\partial x$ gives

$$\frac{dg}{d\varepsilon_k} = -\lambda^{\mathsf{T}} \frac{\partial A}{\partial \varepsilon_k} x, \qquad \frac{dg}{dX} = \lambda^{\mathsf{T}} \frac{\partial b}{\partial X} - \lambda^{\mathsf{T}} \frac{\partial A}{\partial X} x, \tag{5}$$

and in the complex driven case $2\,\mathrm{Re}[\cdot]$ of the same contractions with the transpose solve, since the pencil is complex-symmetric. The element Jacobians are exact, computed by forward-mode dual numbers through the production kernels, and each gradient costs one forward and one adjoint solve on one factorization. Table 6 lists the four modules. Each has a test against a central difference of the full pipeline (perturb, re-assemble, re-solve, re-extract) and mutation tests: the conjugation error A^{H} for A^{T} and a dropped term both fail. In the driven shape case the source $b = \mathrm{i}\omega\mu_0 \int N \cdot J$ depends on the geometry, and omitting $\partial b/\partial X$ fails the check.

For the capacitance $C = \phi^{\mathsf{T}} K(X) \phi$ the stored energy is stationary in the solved potential, so the implicit term vanishes and $dC/dX = \phi^{\mathsf{T}}(\partial K/\partial X)\phi$, the classical self-adjoint result (Park, 2019). The implementation keeps the general adjoint form, and its test measures the implicit part below 10^{-8} of the total. Composed with $E_C = e^2/(2(C_\Sigma + C_J))$, this gives $\partial E_C/\partial X$. On a parallel-plate fixture Newton reaches a target E_C in one step, as expected for its affine parameterization, and a fresh solve reproduces the target to 1.4×10^{-15}. Appendix A applies the same identity, in a native-f64 P2 implementation outside the Burn tape, to a three-parameter electrostatic optimization of the real device. The eigenfrequency derivative of Section 7 is the analogous stationarity result for the eigenproblem.

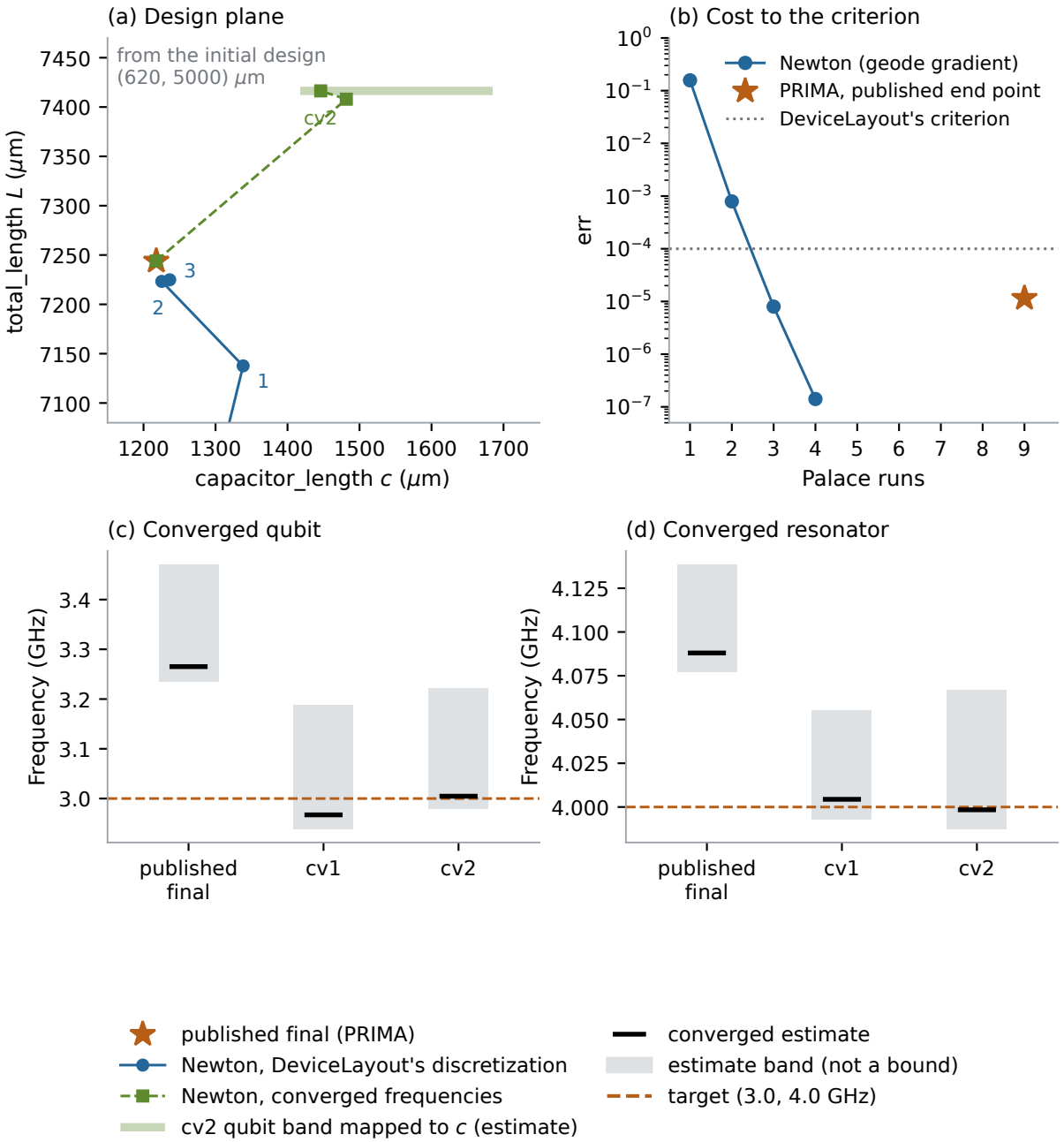


Figure 4: The gradient-assisted re-optimization. (a) The design plane near the optimum: the published final design, the Newton iterates against DeviceLayout's discretization (numbered; the first step starts from the initial design, off the plot), and the two steps against the converged frequencies, with the step-2 qubit band mapped linearly to capacitor_length (an estimate). (b) DeviceLayout's err against the number of Palace runs at its discretization; PRIMA's published end point is plotted at its 9 runs, with err computed from the published frequencies, since its trajectory is not published. (c, d) Converged estimates and extrapolation bands of the published final design and the two converged-target steps, against the targets. Sources: `benchmarks/transmon_devicelayout_optimize/` and `benchmarks/transmon_devicelayout_crosscheck/results.toml`.

Table 7: Artifacts behind each result, relative to the repository root. The generated artifacts are pinned by regression tests.

Result	Artifact
Same-mesh agreement (Table 1)	`benchmarks/transmon_eigen/results.toml`; Palace cells under `reference/fixtures/transmon_palace/`
Cross-check (Table 2, Figure 2)	`benchmarks/transmon_devicelayout_crosscheck/`; Palace cells under `reference/fixtures/transmon_palace/results_crosscheck_1092/`; meshes from `reference/julia/generate_transmon_param.jl`
Capacitance convergence and brackets (Table 3, Figure 3)	`benchmarks/transmon_diffopt/c_sigma_convergence.toml`, `c_sigma_bracket.toml`, `c_sigma_bracket_tensor.toml`, `c_sigma_bracket_anchor_93p95.toml`
Circuit terms L_s and k	`benchmarks/transmon_loop_inductance/results.toml`, `benchmarks/transmon_coupling/results.toml`
Re-optimization (Tables 4–5, Figure 4)	`benchmarks/transmon_devicelayout_optimize/`; Palace cells under `reference/fixtures/transmon_palace/results_optimize_1092/`
Electrostatic design loop (Appendix A)	`benchmarks/transmon_diffopt/refined_optimization/`
Performance (Appendix B)	`benchmarks/transmon_bench_cpu/`, `benchmarks/gpu_driven_scaling/`

Table 8: Evidence tiers. Certified: a guaranteed bound for the meshed polygonal geometry. Measured: a committed solve on a stated discretization. Estimated: an extrapolation with a stated rule, not a bound.

Tier	Results
Certified	C_Σ of X^0: $[77.40, 77.61]$ fF (scalar), $[79.56, 79.76]$ fF (tensor); tensor-minus-scalar gap 2.5–3.0%; the qubit cap of 4.477 GHz, conditional on the circuit model
Measured	Same-mesh agreement (Table 1); Palace at DeviceLayout's discretization within 0.7% of the published pairs; every ladder level; same-mesh agreement on the ladders; the gradient check; 3 Palace runs to DeviceLayout's criterion
Estimated	Converged frequencies and bands (Table 2); the static-route qubit range; L_s and k; the converged design $c \approx 1446\,\mu$m and its +16 to +38% range
Not claimed	Agreement with measurement; a converged hit of 3.0/4.0 GHz; a global optimum; eigenmode-participation (EPR) derivatives; a speed advantage over Palace

9 Reproducibility

Every result in this paper is read from a committed, scripted artifact.[1] Table 7 lists them. The versions are geode-fem commit `3174015` for the same-mesh credential and later commits named in each artifact, Palace `fba6a5b`, DeviceLayout.jl v1.15.0 and Burn 0.21. The DeviceLayout ladders ran on an AWS `r8g.8xlarge` (Graviton4, 32 vCPU); the capacitance brackets ran on a local 28-core, 96 GiB workstation. The Palace runs consume an MSH 2.2 conversion of each mesh, because MFEM rejects DeviceLayout's multi-block MSH 4.1 file; the conversion keeps every node and physical group and reproduces the eigenvalues bit for bit. The figure scripts in `figures/src/` read the artifacts directly.

10 Discussion

What is established, and how. Table 8 sorts the results by the kind of evidence behind them. The brackets are the only bounds. The converged frequencies and the re-optimized design are extrapolations with stated rules and bands, and the static route rests on circuit terms that are themselves extrapolated. The two routes agree to 0.22% on the initial qubit.

[1] Repository URL and archival DOI for the geode-fem source to be added at submission. TODO(operator).

Threats and limitations. Agreement between two codes bounds implementation error, not shared modeling choices, and the 17.49 GHz port-sheet mode is an example of a shared artifact; the rank-1 shunt removes it in geode-fem, and Palace still has it. The extrapolations assume a rate the sheet-edge singularity sets and do not include mesh-family spread, and their high sides are wide. The circuit route uses a bracket certified on a different mesh of the same initial geometry, and L_s and k were extrapolated from two or three discretizations. The re-optimization's transmission-line model rests on one weakly determined δ; it affects the step count, not the iterates' evaluation. The PRIMA comparison counts runs only. The electrostatic certificate is evaluated in f64 and its code is validated by tests, not formally verified. The model is lossless and linear: the junction is $L_J \parallel C_J$, the readout ports are open in the ladders, and the metal is a set of zero-thickness sheets. Only one device is studied. The eigenmode branch is not yet differentiated with respect to participation ratios or Kerr couplings; an interior eigensolve that is robust at the physical shift is the missing piece, and a projected Jacobi–Davidson iteration (Liang et al., 2026) with the eigenpair derivatives of Nelson (1976) is the planned route.

11 Conclusion

The published frequencies of the DeviceLayout.jl SingleTransmon example are reproduced by Palace at the example's own discretization and are 7–9% below the converged qubit frequencies that two independently implemented solvers reach under refinement. A certified bracket on the island capacitance gives the same shift by a static route and caps the initial qubit. A Newton iteration with a geode-fem eigenfrequency gradient in its Jacobian reaches DeviceLayout's criterion in a third of the published Palace runs, and at converged resolution it moves the optimized design to a capacitor about 19% longer, within a band of 16–38%. For design examples that are meant to be copied, the discretization is part of the result, and a solver that supplies gradients and certified bounds makes it cheap to report converged values.

Acknowledgments. geode-fem's development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline: results are traced to committed, human-auditable artifacts, and statements that rest on a scratch probe, on a reading of Palace's source, or on inference are labelled as such.

A Supporting Material: An Electrostatic Design Loop With a Certified Acceptance Test

This appendix reports an electrostatic optimization of the real device whose interest is its acceptance test, not its target. The target, an island capacitance of 93.954 fF, was obtained by reading the published initial-geometry frequency of 4.14 GHz as a design target through the linear junction circuit. Since 4.14 GHz is the frequency of the unmodified initial design, the target has no design meaning; the runs remain a measured example of how much a target reached on an optimization mesh is worth after refinement.

Method. Ten cycles of adaptive refinement turn the committed mesh into a 448,279-tetrahedron optimization mesh. Three junction-pinned harmonic morph fields change the island length, the pad width and the cutout gap. The gradient is the stationary-energy shape derivative of Section 8 for the floating-feedline C_Σ, evaluated with P2 fields and exact quadrature on each affine tetrahedron, in native f64 outside the Burn tape. The problem is underdetermined (three parameters, one target), and the accepted steps are those that reduce the freshly solved residual and keep the minimum element-volume ratio above 0.25. The design is scalar-$\bar{\varepsilon}$.

Result. The optimizer raises C_Σ from 78.387918 to 93.953668 fF in four accepted steps; at the final design all three sensitivities agree with central differences of fully re-solved geometries to 4.16×10^{-7} (rounded up). Seven held-out refinement rounds, none of which feed the optimizer, lower the conforming value to 92.998560 fF at 3,388,197 tetrahedra, and the certificate of Section 6 on the same meshes brackets the converged capacitance of the design in $[92.75, 93.00]$ fF with the scalar model and $[95.34, 95.58]$ fF with the sapphire tensor. The target reached on the optimization mesh is lost by about 1% under refinement, and the certificate bounds the loss. A second run aimed past that bias (target 95.112986 fF) gives $[93.89, 94.14]$ fF scalar and $[96.51, 96.75]$ fF tensor. Palace with second-order elements on the exported adapted mesh agrees with geode-fem's C_Σ to 2.5×10^{-6}, which tests the two assemblers on one discretization and says nothing about the discretization error.

What it shows. A gradient loop converges to the target it is given on the mesh it is given, and a finite-difference check or a fresh solve on that mesh says nothing about the discretization error. The held-out conforming series and the equilibrated-flux certificate together give a two-sided acceptance test about 0.25 fF wide, independent of the optimizer. The same certificate, with the tensor, supplies the bracket of Section 6.

B Performance Context

The timings below qualify the cost of the solvers; none measures the re-optimization or ranks the two codes. The CUDA backend (burn-cuda 0.21) is f32-only today: the underlying cubecl runtime currently disables f64[2] — so every GPU result is mixed-precision-qualified.

CPU eigenmodes. On one otherwise idle EC2 `m6i.4xlarge`, with every request verified to return the six reference modes of Table 1, Table 9 gives median wall seconds and measured core-seconds. Two of the geode-fem requests are tuned using the reference spectrum, and Palace solves about 17% more unknowns from the same input mesh, so these are workload comparisons, not kernel speedups.

[2]Upstream constraint of the burn/cubecl stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

Table 9: Mode-verified CPU timings on one m6i.4xlarge: median wall seconds / measured core-seconds. All rows contain the six reference modes of Table 1. Three repeats per cell except Palace one-rank Save=0 (one run). The two requests tuned on the reference spectrum are marked †. Palace always requests six modes above 4.5 GHz. Source: `benchmarks/transmon_bench_cpu/results_like_for_like_m6i.toml`.

Request	Extra modes	1 wide	8 wide
geode 30 at 4.5 GHz†	24	34.45 / 34.44	29.08 / 40.71
geode 6 at 20 GHz†	0	33.21 / 33.20	27.26 / 38.87
geode port-aware 8 at 4.5 GHz	2	67.26 / 67.25	51.12 / 91.72
Palace, Save=6	0	186.84 / 186.82	33.35 / 266.66
Palace, Save=0	0	113.00 / 112.98	23.64 / 188.89

At eight wide Palace without field output is fastest; at one wide the geode-fem routes take less time, but geode-fem writes no fields and has no counterpart to Palace's error estimator.

Driven solves and memory. On assembled first-order driven problems, geode-fem's AMS preconditioner (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009) reduces one-core solve time relative to Jacobi by 3.4–4.2× across 25,695–897,650 edges, but Palace's AMS is faster at scale (53.3 against 181.3 s at the largest size). On a lumped-port cube family up to 25,695 edges, geode-fem's GPU-f32 matrix-free solve is slower than both assembled CPU configurations at every measured size and reaches parity with CPU matrix-free only at the largest. A geode-fem direct eigensolve at 1,157,564 unknowns completed in 565.5 s at 92.2 GB peak memory, on a different host from Palace's 423.12 s run; the direct factorization's memory, not a controlled crossover, is what that shows. The DeviceLayout ladders of Section 5 used Palace for the largest cells (up to 12.6 million unknowns), and geode-fem's direct solves at $h/2$ peaked at 94 and 136 GB.

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